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RESEARCH ARTICLE

Quantifying Effect Modifier Similarity for Regional Pooling in Multi-Regional Clinical Trials

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Abstract

The ICH E17 guideline describes regional pooling in multi-regional clinical trials (MRCTs) in terms of the similarity of effect modifier distributions but names no measure, and to our knowledge no distance between regional effect modifier distributions has been evaluated for it. We quantify how two regions' distributions of each candidate effect modifier differ by the Wasserstein-1 (W_1) distance in original units. If the conditional average treatment effect is common to all regions, depends only on that modifier and is Lipschitz with clinical slope L_{clinical}, regional average treatment effects differ by at most $\Delta_{\max} = L_{\text{clinical}} W_1$ on a risk- or mean-difference scale; when L_{clinical} is unknown, the smallest slope L^* at which the distance could produce a clinical margin is compared with a sponsor's bound. In a selection simulation (matches supplied), the standardized mean difference stayed at chance up to 2,000 patients per region in some configurations; W_1 and the Kolmogorov–Smirnov (KS) statistic reached the target (AUC 0.9) in every configuration. For true distances of about 0.4 to 1.2 SD (normal reference; equal n), bootstrap coverage from 100 patients per region was 0.92–0.95, below 0.95 beyond Monte Carlo error in 5 of ten cells (0.947–0.951 at the application's operating point). In a GUSTO-I-based example, partner rankings on age and systolic blood pressure differed; a null floor marked distances not distinguishable from no difference. The framework quantifies differences; the pooling judgment remains clinical.

KEYWORDS
multi-regional clinical trial; ICH E17; effect modifier; Wasserstein distance; regional pooling; distributional similarity

1 | INTRODUCTION

Multi-regional clinical trials (MRCTs), conducted across multiple countries or regulatory regions under a single protocol, have become a common design in global pharmaceutical development.[1,2] The International Council for Harmonisation (ICH) E17 guideline, adopted in 2017, established principles for planning and designing MRCTs to evaluate whether treatment effects can be regarded as generalizable across the target population.[3]

Regulatory authorities in Japan and China examine the consistency of treatment effects in regional subpopulations,[4,5] yet individual regional sample sizes in global MRCTs are often insufficient for that purpose. Regional pooling, in which regions with similar patient characteristics are grouped for analysis, is a key strategy for enhancing the assessment of regional consistency, and the ICH E17 guideline describes pooling decisions in terms of the similarity of effect modifier distributions.[3] In the PREVENT trial of secukinumab, a workshop report describes countries with a similar distribution of effect modifiers being clustered post hoc into five pooled regions on binary characteristics (CRP and MRI positivity, sex).[4]

Abbreviations: AMI, acute myocardial infarction; AUC, area under the receiver operating characteristic curve; CATE, conditional average treatment effect; CDF, cumulative distribution function; CI, confidence interval; CRP, C-reactive protein; EDF, empirical distribution function; GUSTO, Global Utilization of Streptokinase and t-PA for Occluded coronary arteries; ICH, International Council for Harmonisation; IPD, individual patient data; KL, Kullback–Leibler; KS, Kolmogorov–Smirnov; MRCT, multi-regional clinical trial; MRI, magnetic resonance imaging; NMPA, National Medical Products Administration; RMSE, root mean squared error; RWE, real-world evidence; SBP, systolic blood pressure; SD, standard deviation; SMD, standardized mean difference.

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An effect modifier is a baseline patient characteristic, such as age, disease severity or a genetic marker, for which the treatment benefit differs across subgroups. Regions with different patient compositions may therefore observe different average treatment effects even if the drug works identically at the individual level, so the similarity of the full effect modifier distribution, not only its average level, matters for regional pooling. We focus on continuous effect modifiers; categorical effect modifiers raise distinct considerations and are outside the present scope.

Despite the regulatory importance of effect modifier distributional similarity, current practice lacks a standardized, clinically calibrated methodology. The ICH E17 guideline provides no specific metric, threshold, or statistical procedure for determining when distributions are "similar enough." Song et al., on the NMPA's implementation of ICH E17, identify the pooling strategy as an area where ambiguity remains and describe clustering populations on distances defined from the correlation between effect modifiers and endpoints.[5] The pooling-strategy chapter of a reference volume on multi-regional trials under ICH E17 introduces, as one of many possible approaches, cluster analysis of regions on region-level summaries of the candidate effect modifiers, and does not evaluate it.[6] Two gaps remain. First, to our knowledge no distance between regional effect modifier distributions has been proposed and evaluated for this purpose: a single summary per modifier leaves two regions that agree on it but differ in spread, shape, or tail behaviour at distance zero, and any finite set of summaries requires deciding in advance which distributional features are relevant. Second, the chapter calls for a judgment of whether differences in the impact of effect modifiers across pooled regions are "clinically meaningful or not" but supplies no metric to operationalize that judgment.

The ingredients are not new in themselves. The inequality on which the clinical calibration rests is the Kantorovich–Rubinstein duality, used in the same form in the optimal-transport analysis of domain adaptation;[7,8] the similarity between a trial sample and a target population has been summarized by propensity-score metrics,[9] a generalizability index[10] and an equivalence test on a trimmed Mallows distance.[11] None of these converts the difference between two regions' distributions of an individual effect modifier into a bound on the regional effect difference whose constant is a clinical slope, or evaluates such a distance for regional pooling. The contribution of this paper is that formulation, the reverse calibration L^* when the slope is unknown, and the evaluation of the estimator's identification and estimation properties.

This paper addresses these gaps by proposing a quantitative framework for assessing effect modifier distributional similarity across regions, based on the Wasserstein-1 (W_1) distance between two cumulative distribution functions, retained in the original measurement units of the effect modifier. We estimate W_1 with bootstrap confidence intervals and link it to clinical interpretation through a calibration framework.

2 | METHODS

Let the conditional average treatment effect (CATE) be denoted $\tau(x) = E[Y(1) - Y(0) \mid X = x]$, where X is the effect modifier. When this function is non-constant, the average treatment effect observed in region r depends on the distribution F_r of the effect modifier in that region:

$$\bar{\tau}_r = \int \tau(x)\, dF_r(x). \tag{1}$$

Assessing pooling suitability therefore requires a measure of distributional difference between regions. Equation (1) presupposes an effect scale on which the regional effect is the average of the conditional effects, that is, a collapsible contrast such as a risk difference, a mean difference or a difference in restricted mean survival time; marginal odds and hazard ratios fall outside Proposition 1 unless the regional effect is first defined by standardization on a collapsible scale.

2.1 | Existing Approaches and Their Limitations

The most widely used measure for comparing covariate distributions across groups is the standardized mean difference (SMD),[12] which by construction captures only differences in location: $N(50, 5^2)$ and $N(50, 15^2)$ yield SMD = 0 despite a threefold difference in spread. Distributional distance measures compare distributions beyond summary statistics: empirical distribution function (EDF) statistics such as the Kolmogorov–Smirnov (KS) statistic, $D_{\mathrm{KS}} = \sup_x |F_1(x) - F_2(x)|$, compare CDFs directly, and information-theoretic divergences such as Kullback–Leibler (KL) divergence take a density-based approach.

Each of these approaches has limitations for comparing effect modifier distributions across regions. EDF-based statistics such as the KS statistic capture full distributional differences but lack a clinically interpretable scale, and the bound on treatment effect heterogeneity they admit is governed by the total variation of the CATE, a constant that is clinically stated only for a

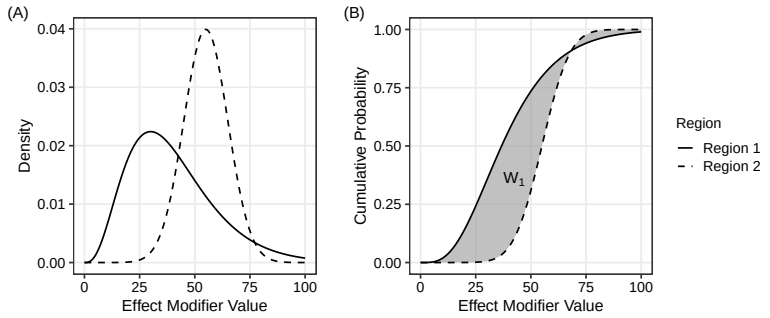


FIGURE 1 The Wasserstein-1 distance as the area between cumulative distribution functions, illustrated with a right-skewed Region 1 (gamma distribution) and a symmetric Region 2 (normal distribution). Panel (A): probability densities. Panel (B): cumulative distribution functions; the shaded area equals $W_1(F_1, F_2)$.

monotone CATE, for which $\mathrm{TV}(\tau) = |\tau(b) - \tau(a)|$ on a support $[a, b]$. KL divergence is asymmetric, infinite for non-overlapping empirical distributions, and requires density estimation that is impractical with small regional samples. These gaps motivate a measure that (i) captures distributional features beyond location, (ii) provides a clinically interpretable scale, and (iii) offers a bound on regional treatment effect differences whose constant is a per-unit clinical slope; the Wasserstein-1 distance and its clinical calibration, developed next, satisfy all three.

2.2 | Per-Effect-Modifier Wasserstein-1 Distance

Among distributional distances that could capture the full distributional differences relevant to regional pooling, we adopt the Wasserstein-1 distance (W_1) for three reasons: it responds to differences beyond location; the heterogeneity bound it admits (Proposition 1 below) has a per-unit clinical slope L_{clinical} as its constant; and that bound is valid for every CATE in the class, sharp when the slope is the only restriction placed on the CATE, and tighter, as a function of W_1 and the mean difference, when the CATE is also monotone; for any particular CATE it is therefore a conservative upper bound. The W_1 distance admits three equivalent representations on the real line.[13,14]

Let F_1 and F_2 denote the cumulative distribution functions (CDFs) of an effect modifier in two regions. Then

$$W_1(F_1, F_2) = \int_{-\infty}^{\infty} |F_1(x) - F_2(x)|\, dx = \int_0^1 |F_1^{-1}(u) - F_2^{-1}(u)|\, du = \inf_{\pi \in \Pi(F_1, F_2)} \int |x - y|\, d\pi(x, y), \tag{2}$$

where F_r^{-1} is the quantile function and $\Pi(F_1, F_2)$ is the set of probability couplings with marginals F_1 and F_2. The CDF form expresses W_1 as the area between the two CDFs (Figure 1); the quantile form as the integrated absolute difference of quantile functions; and the Kantorovich form as the minimum expected absolute displacement to transport one distribution onto the other.[14] Unlike the standardized mean difference, W_1 responds to differences in variance, skewness, and other distributional features.[15]

W_1 carries the units of the underlying effect modifier in each of the three equivalent forms of equation (2), so it is expressed on the original scale and supports direct clinical interpretation: a regional $W_1 = 5$ mmHg in systolic blood pressure means that optimally transporting one distribution onto the other moves probability mass an average distance of 5 mmHg. Values of W_1 for effect modifiers measured on different scales are not directly comparable, so we restrict attention to per-effect-modifier assessment.

W_1 is a proper metric on the space of probability distributions with finite first moment.[13] It also admits a Kantorovich–Rubinstein dual representation[13]

$$W_1(F_1, F_2) = \sup_{\|f\|_{\text{Lip}} \le 1} \left| \int f\, dF_1 - \int f\, dF_2 \right|, \tag{3}$$

which expresses W_1 as the largest possible difference in expected values of a 1-Lipschitz function f between the two distributions ($\|f\|_{\text{Lip}} \le 1$ means $|f(x) - f(y)| \le |x - y|$ for all x, y). If the conditional average treatment effect function $\tau(x)$ is Lipschitz

continuous with constant L_{clinical}, so that $|\tau(x) - \tau(y)| \leq L_{\text{clinical}}|x - y|$, then τ/L_{clinical} is 1-Lipschitz and applying equation (3) yields the central result of this section:

Proposition 1. (Connection to heterogeneity). *Suppose the CATE function τ is the same in every region and depends on the effect modifier X alone. If τ is Lipschitz continuous with constant $L_{clinical}$, then for any two regions with effect modifier distributions F_1 and F_2,*

$$|\bar{\tau}_1 - \bar{\tau}_2| \leq L_{clinical} \cdot W_1(F_1, F_2). \tag{4}$$

Proof. Apply equation (3) with $f = \tau/L_{\text{clinical}}$, which is 1-Lipschitz by assumption. This yields $\left|\int (\tau/L_{\text{clinical}})\, d(F_1 - F_2)\right| \leq W_1(F_1, F_2)$. Combining with equation (1) for $\bar{\tau}_r$ and rearranging yields equation (4). □

The bound is not specific to W_1: for a CATE of bounded variation on a bounded support, integration by parts gives a bound $\text{TV}(\tau) \cdot D_{\text{KS}}$, sharp within its own class, and $W_1 \leq W_2$ gives a valid but looser $L_{\text{clinical}} \cdot W_2$ with the same constant, which, like every valid bound with that constant, is at least $L_{\text{clinical}} W_1$ (Appendix A.2).[13] Equation (4) is sharp when the Lipschitz constant is the only restriction placed on the CATE: by the Kantorovich–Rubinstein duality of equation (3), $L_{\text{clinical}} W_1$ is the supremum of $|\bar{\tau}_1 - \bar{\tau}_2|$ over all common L_{clinical}-Lipschitz CATEs, attained by $\tau^*(x) = L_{\text{clinical}} \int_{x_0}^{x} \text{sgn}(F_2(t) - F_1(t))\, dt$, whose slope changes sign at every crossing of the two CDFs. If τ is also monotone, with $0 \leq \tau' \leq L_{\text{clinical}}$, then $\bar{\tau}_1 - \bar{\tau}_2 = \int (F_2 - F_1)\, \tau'\, dx$ lies between $-L_{\text{clinical}} A_-$ and $L_{\text{clinical}} A_+$, where $A_\pm = \int (F_2 - F_1)_\pm\, dx$ satisfy $A_+ + A_- = W_1$ and $A_+ - A_- = \mu_1 - \mu_2$, so that

$$\sup_{\tau \text{ monotone, } L_{\text{clinical}}\text{-Lipschitz}} |\bar{\tau}_1 - \bar{\tau}_2| = L_{\text{clinical}} \max(A_+, A_-) = \tfrac{1}{2} L_{\text{clinical}} \left(W_1 + |\mu_1 - \mu_2|\right), \tag{5}$$

which equals $L_{\text{clinical}} W_1$ when the CDFs do not cross and $L_{\text{clinical}} W_1/2$ when the means coincide; $\Delta_{\max} = L_{\text{clinical}} W_1$ is therefore a conservative upper bound by construction.

Because the bound involves W_1 in its original measurement units multiplied by a clinical slope, no further normalization is required and we adopt W_1 in its original units throughout.

2.3 | Estimation

Given samples $\{X_{1,i}\}_{i=1}^{n_1}$ from region 1 and $\{X_{2,j}\}_{j=1}^{n_2}$ from region 2, the Wasserstein-1 distance is estimated using empirical distribution functions. Since $\hat{F}_1$ and $\hat{F}_2$ are piecewise constant (right-continuous step functions), the integral in equation (2) reduces to a finite sum over the intervals between consecutive distinct values of the pooled sample:

$$\widehat{W}_1 = \sum_{k=1}^{m-1} |\hat{F}_1(x_{(k)}) - \hat{F}_2(x_{(k)})| \cdot (x_{(k+1)} - x_{(k)}), \tag{6}$$

where $\hat{F}_r$ denotes the empirical CDF of region r and $x_{(1)} < \cdots < x_{(m)}$ are the m distinct values of the pooled sample. When $\{x : F_1(x) = F_2(x) \in (0, 1)\}$ has Lebesgue measure zero, as when the two CDFs cross only at isolated points (Appendix A.1, with its moment and sample-ratio conditions), the limit is Gaussian and the ordinary nonparametric bootstrap is consistent;[16] where the CDFs coincide on an interval, including $F_1 = F_2$, the naive bootstrap is not consistent.[17,16,18] A Wald interval is governed by the same condition and requires an estimate of the limit's variance, which depends on $\text{sgn}(F_1 - F_2)$ and hence on where the CDFs cross, whereas the percentile bootstrap requires none; we therefore employ the nonparametric percentile bootstrap with $B = 2{,}000$ replicates.[19] Each replicate resamples each region independently, with replacement and at its own size, and the 95% interval is formed by the 2.5th and 97.5th sample percentiles of the B replicates (R's default quantile definition, type 7).

Because W_1 is a distance, $\widehat{W}_1$ is non-negative by construction and typically positive when the two regions are draws from the same distribution: it is biased upward at every sample size, $E\,\widehat{W}_1 \geq W_1$ (Appendix A.1), and the percentile interval does not correct this bias. We refer to the sampling distribution of $\widehat{W}_1$ under $F_1 = F_2$ as the null floor; it is obtained, per region pair, by drawing two independent resamples, of the two regions' actual sizes, from one region's empirical distribution—the anchor's in Section 4.2—and recomputing $\widehat{W}_1$. An observed $\widehat{W}_1$ at or below its 95th percentile is one the estimator would produce with appreciable probability from identical distributions, so it is not evidence of a difference; how large the difference could be is read from the upper confidence limit of $\widehat{W}_1$. Declaring a distance unresolved in this way is a one-sided test of $F_1 = F_2$ at level 0.05; the comparisons of Section 4 are not adjusted for multiplicity, since no inference about pooling is drawn from the number of unresolved distances.

2.4 | Interpretation and Clinical Calibration

We refer to the Lipschitz constant of the CATE function $\tau(x)$ as the clinical slope L_{clinical}: the maximum change in the treatment effect per unit change in the effect modifier, on the outcome scale of the trial—for example, absolute 30-day mortality per year of age or per mmHg of systolic blood pressure, quoted as a proportion, so that $L_{\text{clinical}} = 1 \times 10^{-2}$ per year is one percentage point per year and a margin Δ_{clin} stated in percentage points is divided by 100. Meta-regression on treatment-by-covariate interaction in prior individual patient data (IPD) estimates the average interaction slope,[20,21] which is, in absolute value, a lower bound on the Lipschitz constant, so an L_{clinical} specified from it should carry a sponsor-specified safety factor.

From Proposition 1, the maximum potential difference in regional average treatment effects attributable to effect modifier distributional differences is

$$\Delta_{\max} = L_{\text{clinical}} \cdot W_1(F_1, F_2), \tag{7}$$

expressed in the units of the trial outcome scale; when L_{clinical} is independently known, equation (7) provides a direct clinical-scale translation of the observed distributional difference.

The estimator $\widehat{W}_1$ has a percentile bootstrap CI (Section 2.3), and inferential uncertainty propagates linearly to $\Delta_{\max}$ when L_{clinical} is treated as fixed: $[\Delta_{\max,L}, \Delta_{\max,U}] = L_{\text{clinical}} \cdot [W_{1,L}, W_{1,U}]$. When L_{clinical} itself is uncertain or unavailable, we solve equation (7) for L_{clinical} at a pre-specified clinical margin Δ_{clin} (the minimal clinically important treatment effect difference, the non-inferiority margin, or the targeted overall effect), yielding the required clinical slope

$$L^* = \frac{\Delta_{\text{clin}}}{W_1(F_1, F_2)}. \tag{8}$$

L^* is the smallest clinical slope under which the observed distributional difference could produce a clinically meaningful regional treatment effect difference, within the class of CATEs restricted only by their slope. Because L^* is a decreasing function of W_1, the bootstrap CI of $\widehat{W}_1$ gives the percentile interval $[\Delta_{\text{clin}}/W_{1,U},\ \Delta_{\text{clin}}/W_{1,L}]$ for L^*. If a plausible upper bound on L_{clinical} from prior evidence (denoted L_{UB}) lies below L^*, the distributional difference cannot produce such a difference, provided $L_{\text{clinical}} \leq L_{\text{UB}}$.

Specifying L_{clinical} (or its upper bound L_{UB}) is a clinical judgment, informed but not determined by prior evidence, for which the sponsor carries responsibility—the same footing as a non-inferiority margin. A prognostic gradient alone guarantees no bound on the modification slope, prognostic and treatment-effect gradients being distinct quantities.[22,23]

The framework is deliberately estimation-centered rather than testing-based: W_1 has no universal cutoff that separates "similar" from "dissimilar," and what a given W_1 means clinically depends on the clinical slope L_{clinical}, which is external to the distributional comparison. Per-pair $\Delta_{\max}$ (or L^*) with bootstrap CIs are inputs for sponsor and regulatory judgment, which is not driven by a significance test;[24] the one test, the null floor of Section 2.3, governs how an estimate is displayed and read, not whether regions are pooled.

3 | SIMULATION STUDIES

Study 1 evaluates identification: whether each distance measure can detect, at some sample size, that a candidate region's effect modifier distribution differs from the anchor's, on a selection task in which candidates built from the anchor's population are ranked against candidates built from a different population, with the number of true matches supplied. It does not order regions that all differ from the anchor, nor rank them by effect difference, which depends on the unknown CATE: under a linear CATE only the mean difference matters, whereas under the V-shaped CATE $\tau(x) = L_{\text{clinical}}|x - 50|$ a scale difference about the centre 50 produces the full $L_{\text{clinical}} W_1$ while the SMD is zero. Study 2 evaluates bias, variability and bootstrap coverage of $\widehat{W}_1$.

3.1 | Study 1: Identification

3.1.1 | Design

Study 1 follows the ADEMP structure for simulation studies.[25] Four worlds are simulated, each with regions constructed either from the anchor's own population (true matches) or from a discordant one. Set 1 (Gaussian) places discordance in location and scale; Set 2 (log-normal) in location and in dispersion combined with shape; Set 3 (mixture) in distributional shape alone,

every region, matched or discordant, having mean 50 and SD 10. Set 4 (extremes) places a 5% subgroup in each tail, 40 units from the centre in the anchor; discordant regions displace both subgroups (to 70 or 100 units), raise their prevalence to 12% (mean fixed), or displace only the upper subgroup to 100 units, and two control regions instead shift their whole population by 2 and 3 units. In the three Set 4 cells in which a subgroup is displaced (symmetric and asymmetric severity) or its prevalence raised, a subgroup of prevalence ε displaced by d changes the KS statistic by at most about ε whatever d is, but W_1 by εd, so W_1's sample-size advantage there follows from the construction; the bulk-shift control, a location shift of the whole population, removes it. Ground truth is the construction label, so the target is defined without reference to any of the compared measures and the comparison is not circular.

Three measures enter the comparison: W_1; the KS statistic, the most common comparison of full distributions; and SMD, the standard comparison of location. Other distances, tasks and performance measures run under the protocol (representative-value distances, a log-scale SMD, a harm measure based on the largest W_1 among the retained regions, the probability that the retained set includes a discordant region, a threshold-rule study, an eligibility-rule sensitivity analysis, and a clustering task whose registered criterion for estimating the number of groups was not met) are not reported; this paper compares distribution-level measures with the single-summary SMD and prescribes no grouping procedure; the outputs are in the repository.

An anchor and nine candidate regions are drawn; each measure ranks the candidates by their distance to the anchor, and the k closest are retained, k being the number of true matches (an oracle the application does not use). Performance is the area under the receiver operating characteristic curve (AUC) for discriminating true matches from discordant candidates (chance = 0.5), over 10,000 replications per cell at the base grid (n = 25–100 per region) and 3,000 at the extended grid (n = 150–2,000). The extended grid, the accuracy target and the definitions of the two behaviours below were fixed after the base-grid results had been seen, and the Monte Carlo intervals of Table 1 were added later; the reported target is the middle of the three examined.

3.1.2 | Two Behaviours: Blind and Recovering

Extending the sample-size grid to n = 2,000 separates the measures into two behaviours (Figure 2). A measure is blind in a cell when its AUC stays at chance through n = 2,000: the population value of its distance carries no information about the construction label, so the task is unsolvable for that measure at any sample size, an identification failure and not a power failure. This is exact by construction in every mean-matched cell, where the population SMD is zero. On the selection task, SMD is blind in 7 of the 13 cells (marked in Table 1; AUC 0.49–0.51 across the full grid in all of them).

A measure is recovering when its AUC reaches the target and stays above it at every larger sample size: identification succeeds, and what remains is a required-sample-size question. W_1 and the KS statistic are recovering in every cell; the KS statistic is blind nowhere, reaching selection AUC 0.98 at n = 2,000 even in its weakest cell, asymmetric severity in Set 4.

3.1.3 | Selection: Required Sample Sizes

Table 1 reports, for all 13 cells, the per-region sample size at which each measure's selection AUC reaches 0.9, with 95% Monte Carlo intervals. The intervals place W_1 below the KS statistic in 11 cells, three of them the Set 4 subgroup cells, where the advantage follows from the construction (Section 3.1.1), and the others including all six cells of Sets 1 and 2, which contain no outlying subgroup; they place the KS statistic below W_1 in 1, the bulk-shift control, and overlap in 1, the shape-symmetric mixture. For the bulk-shift control, a check registered after an exploratory look (protocol, with the prior look disclosed, and outputs in the repository) found that at n = 200 per region the Set 4 anchor population multiplied the null sampling standard deviation of $\widehat{W}_1$ by 1.62 (Monte Carlo SE 0.04) relative to $N(50, 10^2)$, left that of the distribution-free KS statistic unchanged (1.00, SE 0.02), and reversed the sign of the paired AUC difference between W_1 and the KS statistic (0.029, SE 0.002, under the Gaussian anchor; -0.032, SE 0.004, under the Set 4 anchor).

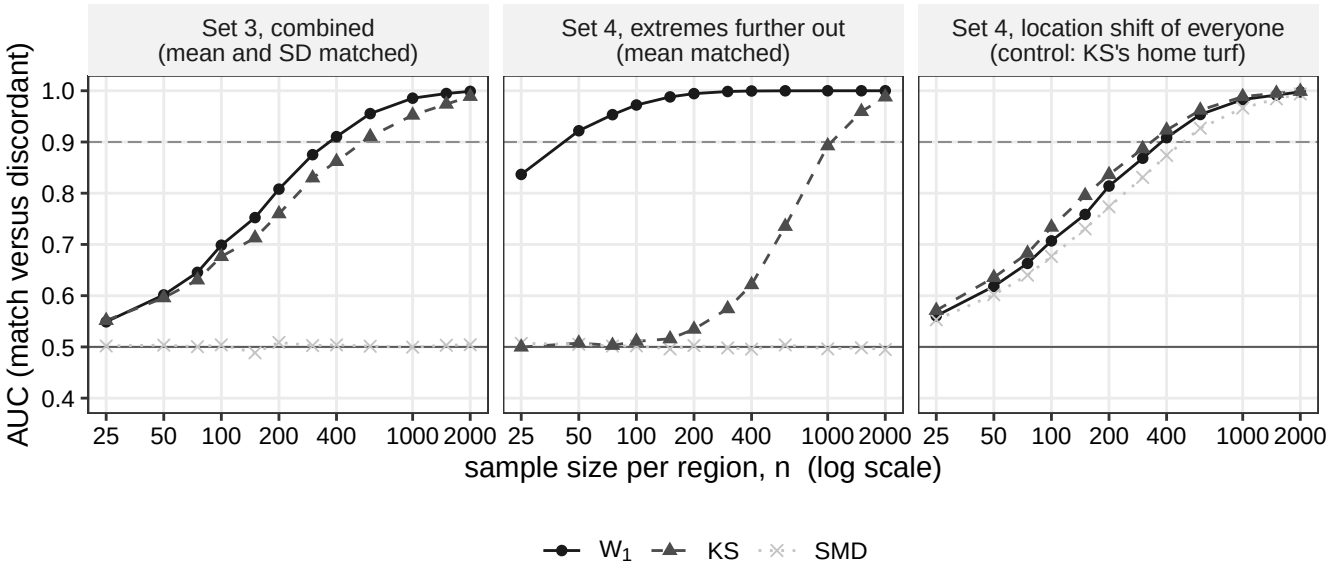


FIGURE 2 Selection AUC (true matches versus discordant candidates) as a function of per-region sample size (n = 25–2,000, log scale) for three distance measures in three cells of Study 1: the Set 3 combined cell (mean- and SD-matched mixtures), the Set 4 symmetric-severity cell (mean-matched displaced extremes), and the Set 4 bulk-shift control cell (uniform location shift of the whole population). The dashed horizontal line marks AUC = 0.9; chance corresponds to 0.5.

TABLE 1 Per-region sample size at which selection AUC reaches 0.9, by cell and measure, interpolated on $\log n$ over the grid n = 25–2,000, with 95% Monte Carlo intervals from the delta method applied to the log-n interpolation. Cells whose AUC stays at chance (0.49–0.51) across the full grid are marked "blind"; ≤ 25 marks a target reached at the smallest grid size, where no interval is available, and such a cell counts as separated when 25 lies below the other measure's lower limit.

Cell	W_1 [95% MC interval]	KS [95% MC interval]	SMD [95% MC interval]	W_1 vs KS
Set 1, location	33 [31, 33]	39 [37, 40]	30 [29, 31]	separate (W_1 fewer)
Set 1, scale	≤ 25	76 [73, 78]	blind	separate (W_1 fewer)
Set 1, combined	35 [33, 36]	62 [60, 63]	67 [64, 68]	separate (W_1 fewer)
Set 2, location	129 [123, 135]	157 [149, 165]	134 [127, 140]	separate (W_1 fewer)
Set 2, shape	32 [30, 33]	77 [75, 79]	blind	separate (W_1 fewer)
Set 2, combined	60 [58, 62]	188 [182, 193]	529 [509, 548]	separate (W_1 fewer)
Set 4, symmetric severity	42 [41, 43]	1,047 [1,011, 1,083]	blind	separate (W_1 fewer)
Set 4, symmetric prevalence	98 [94, 101]	367 [356, 378]	blind	separate (W_1 fewer)
Set 4, asymmetric severity	67 [64, 69]	1,252 [1,226, 1,278]	530 [507, 554]	separate (W_1 fewer)
Set 4, bulk shift (control)	377 [362, 392]	333 [318, 348]	489 [469, 509]	separate (KS fewer)
Set 3, combined	368 [354, 381]	550 [528, 572]	blind	separate (W_1 fewer)
Set 3, shape-skewed	537 [520, 554]	905 [865, 945]	blind	separate (W_1 fewer)
Set 3, shape-symmetric	690 [661, 718]	716 [687, 747]	blind	overlap

Cells are contrast types within each world; the "combined" cell holds discordant regions that differ from the anchor in both of the world's contrast types at once (in Set 3, asymmetric and bimodal), not a pool of the other cells.

Intervals reflect only the Monte Carlo error of the two AUC estimates that bracket the crossing, not the error of interpolating between grid points; limits are rounded outward to whole patients. "Separate" means the W_1 and KS intervals do not overlap. All measures are computed on the same replications, so non-overlap is a conservative indication of a difference and overlap does not show equal performance.

3.2 | Study 2: Estimation Properties

3.2.1 | Design

We designed seven systematic scenarios for methodological validation, examining controlled distributional differences in location, scale, scale combined with skewness (S6, a mean-matched log-normal), and a location–scale combination. Table 2 summarizes these scenarios with their true W_1 values, in the original units of the effect modifier.

TABLE 2 Study 2 simulation scenarios with true W_1 values.

ID	Description	Distribution 1	Distribution 2	True W_1
S1	Null (identical)	$N(50, 10^2)$	$N(50, 10^2)$	0.00
S2	Location 0.2σ	$N(50, 10^2)$	$N(52, 10^2)$	2.00
S3	Location 0.5σ	$N(50, 10^2)$	$N(55, 10^2)$	5.00
S4	Location 1.0σ	$N(50, 10^2)$	$N(60, 10^2)$	10.00
S5	Scale $1.5\times$	$N(50, 10^2)$	$N(50, 15^2)$	3.99
S6	Skew + scale (Log-normal)	$N(50, 10^2)$	$\text{LogN}(3.787, 0.5^2)$	12.16
S7	Location + Scale	$N(50, 10^2)$	$N(55, 15^2)$	5.83

S6: $\sigma_{\log} = 0.5$ and $\mu_{\log} = \log 50 - \sigma^2_{\log}/2 = 3.787$, giving mean 50 and SD 26.6. True W_1 is exact for S1–S5 ($\sqrt{2/\pi}\,|\sigma_1 - \sigma_2|$ in S5) and obtained by numerical quadrature for S6 and S7; the performance measures use the exact values 3.989423 (S5), 12.159806 (S6) and 5.833155 (S7).

For each scenario, we generated samples of size n = 50, 100, and 200 per region. We performed 10,000 replications per scenario–sample size combination; the Monte Carlo standard error is at most 0.005 for any reported proportion, and over the 21 cells it is at most 0.026 for bias, 0.021 for RMSE and 0.024 for mean CI width. Bootstrap confidence intervals were computed as in Section 2.3 with B = 2,000 resamples; the performance measures are defined in the note to Table 3.

3.2.2 | Results

Figure 3 and Table 3 report bias, coverage and mean CI width of $\widehat{W}_1$ (the table also RMSE), in the original units of the effect modifier. The upward bias of Section 2.3 is largest under the null (S1: 2.46 at n = 50, 1.26 at n = 200), where $F_1 = F_2$; for S3–S7 it was small relative to the true W_1 and at n = 200 at most 0.193 (S7).

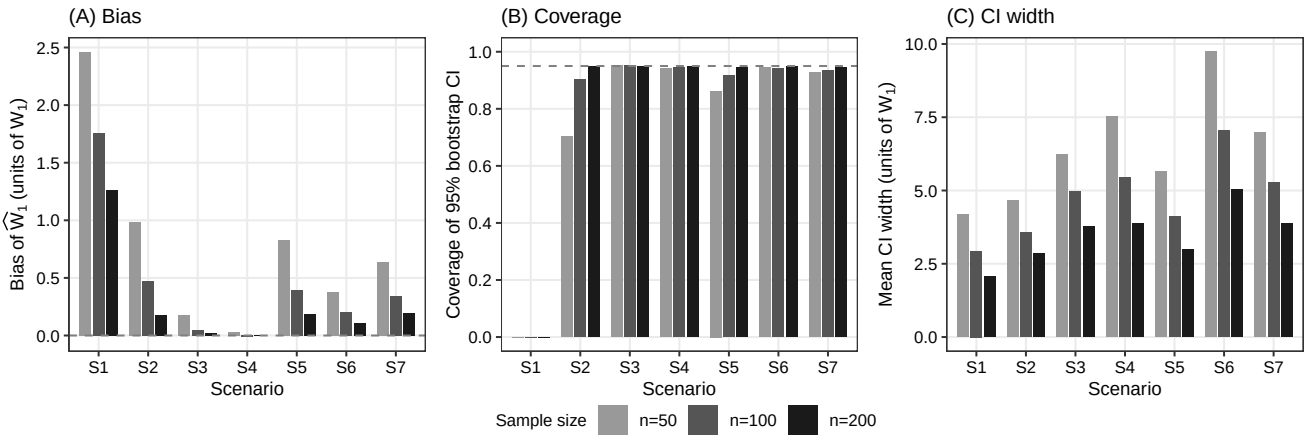


FIGURE 3 Bias (panel A), coverage of the 95% percentile bootstrap CI (panel B) and mean CI width (panel C) of $\widehat{W}_1$ by scenario (S1–S7) and per-region sample size (n = 50, 100, 200), in the original units of the effect modifier. Dashed lines mark zero bias and the nominal 0.95; S1 coverage is zero (Section 3.2.2).

TABLE 3 Bias, coverage of the 95% percentile bootstrap confidence interval, RMSE and mean CI width of $\widehat{W}_1$ by scenario and sample size, in the original units of the effect modifier.

Scenario	True W_1	n	Bias	Coverage	RMSE	Mean CI width
S1 (Null)	0.00	50	2.463	0.000	2.63	4.19
		100	1.759	0.000	1.87	2.94
		200	1.260	0.000	1.34	2.07
S2 (0.2σ)	2.00	50	0.989	0.703	1.60	4.67
		100	0.472	0.904	1.13	3.57
		200	0.174	0.947	0.85	2.84
S3 (0.5σ)	5.00	50	0.175	0.951	1.83	6.22
		100	0.051	0.951	1.37	4.96
		200	0.021	0.948	0.98	3.77
S4 (1.0σ)	10.00	50	0.034	0.942	1.98	7.52
		100	−0.013	0.945	1.42	5.45
		200	0.005	0.948	1.00	3.89
S5 (Scale)	3.99	50	0.830	0.862	1.64	5.65
		100	0.393	0.917	1.11	4.10
		200	0.187	0.946	0.77	2.97
S6 (Skew)	12.16	50	0.377	0.945	2.61	9.73
		100	0.203	0.942	1.86	7.04
		200	0.111	0.947	1.29	5.03
S7 (Loc+Scale)	5.83	50	0.638	0.928	2.00	6.97
		100	0.340	0.935	1.45	5.27
		200	0.193	0.943	1.01	3.87

Bias is Mean$(\widehat{W}_1) - W_1$ and RMSE is $\sqrt{\text{Mean}[(\widehat{W}_1 - W_1)^2]}$; coverage is the proportion of the 95% percentile bootstrap intervals that contain W_1, and is zero under S1 because $W_1 = 0$ lies at the parameter space boundary and a percentile interval cannot contain it; CI width is the mean of (upper − lower). Scenarios are specified with $\sigma = 10$. Results based on 10,000 replications with $B = 2{,}000$ bootstrap resamples.

For S3–S7 at $n \geq 100$, coverage ranged over 0.92–0.95; of these ten cells, 5 lie more than two Monte Carlo standard errors (at most 0.005) below 0.95, the lowest being S5 at $n = 100$ (0.917). The shortfall is largest in S5 and S7, whose CDFs cross; there the intervals lie above the true value, in line with the upward bias of Section 2.3, whereas in S6 and in S4 at $n = 100$, where the bias is smaller relative to the sampling standard deviation of $\widehat{W}_1$, the shortfall is small and the misses fall on both sides. At $n = 50$ coverage was lowest in S2, near the boundary (0.703), and in S5 (0.862).

Coverage under S1 is zero because its true W_1 lies at the parameter space boundary: with continuous data every bootstrap replicate of $\widehat{W}_1$ is positive, so a percentile interval cannot contain zero at any sample size. Whether an observed distance is distinguishable from no difference is read against the null floor, whose 95th percentile under S1 is 4.21, 2.97 and 2.11 at $n = 50$, 100 and 200 ($\sigma = 10$, equal n). Where coverage falls short (S2, S5 and S7 at smaller n), the interval misses from below: the true value exceeded the upper limit in at most 1.1% of replications at any sample size (never in S2), against 2.5% nominal, so the upper confidence limit is a conservative reading of how large a difference could be in these scenarios; not in every scenario whose CDFs cross, since in S6 the true value exceeded the upper limit at about the nominal rate.

Because the application reads an unresolved distance at its upper limit, the one-sided coverage of the 97.5% upper limit, $P(W_1 \leq U)$, is the operating characteristic that matters there: for S3–S7 at $n \geq 100$ it ranged over 0.9710–0.9999 against a nominal 0.975, above nominal where the interval lies above the truth (S5, S7, and S3 at $n = 100$), near nominal in S6 and in S3 at $n = 200$, and below it in S4 at $n = 100$. A basic interval, examined post hoc, exceeded the percentile interval's coverage only under S1 and in S2 at $n = 50$ (0.809 against 0.703; repository).

A registered extension of Study 2 (protocol and outputs in the repository) added cells outside the original design: unequal regional sizes, a heavy-tailed reference distribution, integer-grid data, $n = 1{,}000$ per region and the application's operating point (Table 4). Across the ten non-null cells coverage ranged over 0.921–0.952 (Monte Carlo standard error at most 0.005; 2000–10,000 replications per cell); at the application's operating point it was 0.951 with equal sizes and 0.947 with unequal, rounded samples, with upper-limit coverage 0.9825 and 0.9775.

For S3 on the age scale with the illustrative bound $L_{\text{UB,age}} = 0.01$/yr of the GUSTO-I application (Section 4.1), a $\widehat{W}_1$ CI width of 4.96 years at $n = 100$ corresponds to a $\Delta_{\max}$ interval about 5 times the 1%pt margin used in Section 4.4, so $n = 100$ per region does not by itself make the $\Delta_{\max}$ interval narrow relative to a clinical margin.

TABLE 4 Registered extension of Study 2: unequal regional sizes, a t_3 reference distribution, integer-grid data, n = 1,000 per region and the application's operating point (0.2 SD). Bias, RMSE and mean width in the original units; coverage of the 95% percentile interval and of its one-sided 97.5% upper limit.

Cell	n_1	n_2	Reps	True W_1	Bias	RMSE	Coverage	Upper-limit coverage	Mean width
E1 unequal n	100	300	10,000	5.00	0.056	1.14	0.947	0.9784	4.25
E1 unequal n	200	1,000	5,000	5.00	0.017	0.78	0.941	0.9734	2.97
E2 heavy tails: t_3, null	100	100	10,000	0.00	1.958	2.09	0.000	1.0000	3.08
E2 heavy tails: t_3, shift	100	100	10,000	5.00	0.491	1.28	0.921	0.9947	4.44
E3 integer grid	100	100	10,000	5.00	0.064	1.38	0.945	0.9779	4.97
E3 integer grid	200	200	10,000	5.00	0.021	0.98	0.950	0.9789	3.77
E3 integer grid	1,000	1,000	2,000	5.00	0.019	0.45	0.952	0.9785	1.74
E4 large n: S3	1,000	1,000	5,000	5.00	−0.002	0.45	0.946	0.9746	1.74
E4 large n: S5	1,000	1,000	5,000	3.99	0.034	0.35	0.944	0.9826	1.35
E5 small distance	1,000	1,000	2,000	2.00	0.034	0.44	0.951	0.9825	1.66
E6 small distance, unequal n, rounded	2,900	1,200	2,000	2.00	−0.001	0.35	0.947	0.9775	1.37

Cells: E1, unequal regional sizes (100 against 300; 200 against 1,000); E2, a t_3 reference scaled to SD 10, under the null and under a 5-unit shift; E3, integer-grid data (the S3 contrast rounded to integers at n = 100, 200 and 1,000); E4, n = 1,000 per region for the S3 and S5 contrasts; E5 and E6, the application's operating point (the 0.2 SD contrast of S2 at n = 1,000 per region, and at 2,900 against 1,200 with both samples rounded to multiples of 10, as systolic blood pressure is recorded).
Under the t_3 null the true W_1 is zero, so percentile coverage is structurally zero and upper-limit coverage identically 1, as under S1. The rounded-data truth is the W_1 of the rounded population distributions; Monte Carlo standard error of coverage at most 0.005. Registered (`PROTOCOL_study2_extension.md`) after the external review; E5, E6 and the integer-grid cell at n = 1,000 were added by amendment before being run.

4 | APPLICATION: HYPOTHETICAL THROMBOLYTIC MRCT USING GUSTO-I DATA

We illustrate the per-effect-modifier Wasserstein-1 framework through a hypothetical scenario grounded in publicly available data from GUSTO-I, a large international trial in acute myocardial infarction (AMI). A sponsor planning a Phase 3 MRCT of a novel thrombolytic agent (Drug T) uses IPD from GUSTO-I (N = 40,830; 16 regions)[26] to quantify how the distributions of candidate effect modifiers in other regions differ from those in a designated anchor region, treating the GUSTO-I regions as stand-ins for the regional populations the new trial would enrol; GUSTO-I was not designed as an MRCT and its regions are anonymized.

4.1 | Scenario Design and Effect Modifier Selection

The sponsor takes age and systolic blood pressure (SBP) as the candidate effect modifiers for Drug T. The Fibrinolytic Therapy Trialists' (FTT) meta-analysis of existing fibrinolytic agents[27] concluded that, among patients presenting with ST elevation or bundle-branch block, benefit is present irrespective of age and of blood pressure, and reports no per-unit CATE slope on the absolute mortality scale for either; we therefore treat L_{clinical} as unknown for both and apply the L^* pathway (equation 8). The illustrative bounds are $L_{\text{UB,age}} = 1 \times 10^{-2}$ per year and $L_{\text{UB,SBP}} = 2 \times 10^{-3}$ per mmHg on the 30-day mortality scale, set so that one year of age or 5 mmHg of SBP would reach a margin Δ_{clin} of 1 percentage point; the FTT meta-analysis[27] and the GUSTO-I prognostic model[28] are reference points, not sources of the bounds.

The sponsor selects Region 8 as the anchor region (n = 2,916 patients). Records with SBP = 0 mmHg (6 patients) or age > 100 years (3 patients) were retained; excluding them changed no null-floor classification below and 2 adjacent SBP ranks, and winsorizing at the 1st and 99th percentiles changed 4 of the 30 ranks; both checks were made post hoc. SBP is heaped on multiples of 10 mmHg, most in Region 8 (79.5% of readings), so recording practice may contribute to the SBP distances (not quantified here).

4.2 | Null Floor: Which Distances Are Distinguishable from No Difference?

We computed the null floor of Section 2.3 separately for each candidate partner, from two independent resamples, with replacement, of Region 8's (R8) empirical distribution at sizes n_{R8} = 2,916 and n_{partner}, recomputing $\widehat{W}_1$ over 2,000 repetitions. We call an observed distance unresolved when it lies at or below the 95th percentile of its null distribution (a one-sided test of $F_1 = F_2$ at level 0.05). A permutation test of $F_1 = F_2$ on the pooled sample classifies all 30 distances in the same way (30 of

30 agree). On age, the distances of six of the fifteen partners—R1, R4, R5, R7, R9 and R15—are unresolved at level 0.05; on systolic blood pressure, fourteen of fifteen are resolved, the single exception being R2.

How large the differences of the six partners unresolved at level 0.05 could be is read from the upper confidence limits (Figure 4, Table 5), which for small distances near the boundary err above rather than below the true value in Study 2 (Section 3.2.2). The GUSTO-I distances are 0.03–0.30 SD of Region 8's distributions, below the 0.4 SD at which S3–S7 begin and in places above the 0.2 SD of S2; the extension cells at 0.2 SD, the application's operating point (Table 4), gave percentile coverage 0.951 and 0.947 and upper-limit coverage 0.9825 and 0.9775.

4.3 | Distributional Assessment: Per-Effect-Modifier W_1 Across 15 Partner Regions

We computed $\widehat{W}_1$ between Region 8 and each of the remaining 15 regions for both candidate effect modifiers, with percentile bootstrap 95% confidence intervals (B = 2,000): Figure 4 orders the partners by $\widehat{W}_1$ and Table 5 lists them by region number with their ranks.

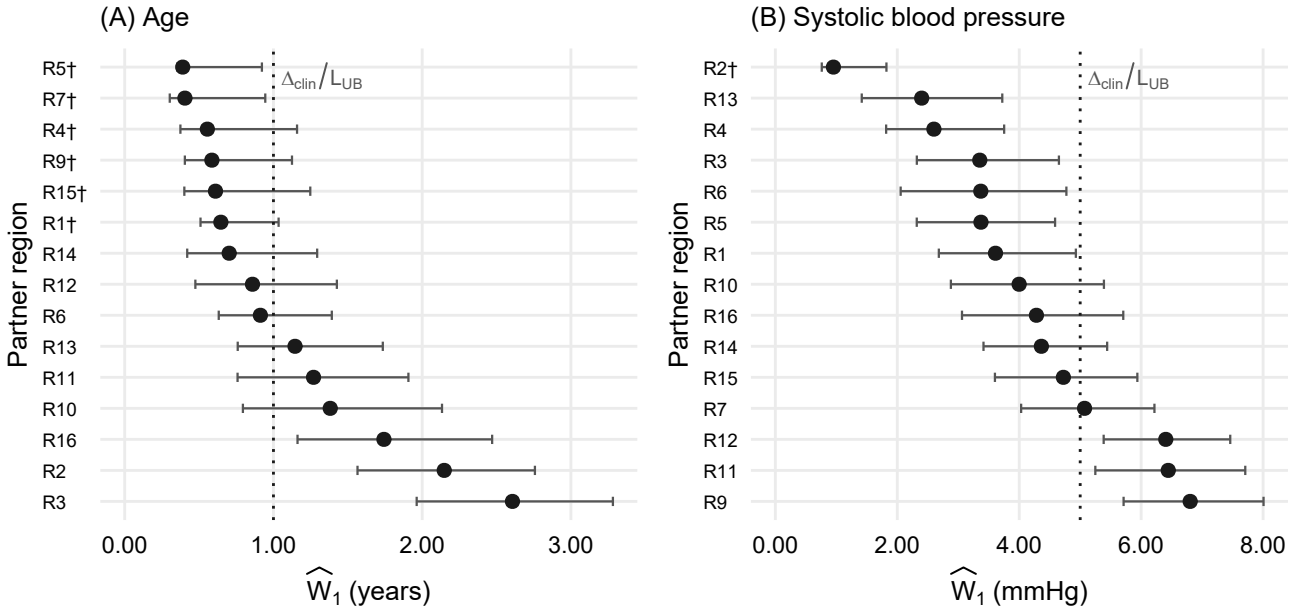


FIGURE 4 $\widehat{W}_1$ with 95% percentile bootstrap CIs for Region 8 versus each partner region (GUSTO-I), ordered by $\widehat{W}_1$ within each panel, in original units: panel (A) age (years), panel (B) systolic blood pressure (mmHg). †Distance unresolved against its null floor (Section 4.2). The dotted vertical line marks $\Delta_{\text{clin}}/L_{\text{UB}}$, the distance at which the illustrative bound of Section 4.1 would produce Δ_{clin} = 1%pt: 1 year for age and 5 mmHg for SBP. The line depends on the sponsor's illustrative bound and is not a threshold for pooling.

Several partners fall in the middle of each ranking, where a rank is not resolved against its neighbours: the rank intervals of Table 5, obtained by resampling patients within every region, show that R6, ranked ninth on age and fifth on SBP, can move across most of the middle of the table, whereas the most distant partners, R2 on age and R9 on SBP, have rank intervals confined to the most distant positions, so a bare rank is not read as a resolved order. R2 has the second-largest $\widehat{W}_{1,\text{age}}$ (2.15 years) but the smallest $\widehat{W}_{1,\text{SBP}}$ (0.95 mmHg), while R9 is among the six smallest $\widehat{W}_{1,\text{age}}$ (0.59 years) but the largest $\widehat{W}_{1,\text{SBP}}$ (6.80 mmHg). Both small distances are unresolved against their null floors (Section 4.2), so the reversal rests on the two large ones and illustrates that the distance on one candidate effect modifier need not carry information about the distance on another.

TABLE 5 Per-effect-modifier Wasserstein-1 distance $\widehat{W}_1$ in original variable units (years for age, mmHg for SBP) with 95% percentile bootstrap CIs for Region 8 versus each partner region (GUSTO-I), listed by region number, with each partner's rank by $\widehat{W}_1$ within each candidate effect modifier.

Partner	n	$\widehat{W}_{1,\text{age}}$ (yr) [95% CI]	Rank	$\widehat{W}_{1,\text{SBP}}$ (mmHg) [95% CI]	Rank
R1	2188	0.65 [0.51, 1.03]	1–6† [2, 10]	3.61 [2.68, 4.93]	7 [3, 10]
R2	2952	2.15 [1.57, 2.76]	14 [13, 15]	0.95 [0.76, 1.82]	1† [1, 2]
R3	2030	2.61 [1.96, 3.28]	15 [14, 15]	3.35 [2.32, 4.65]	4 [2, 9]
R4	2876	0.56 [0.37, 1.16]	1–6† [1, 9]	2.60 [1.82, 3.75]	3 [2, 6]
R5	1909	0.39 [0.35, 0.92]	1–6† [1, 9]	3.37 [2.32, 4.59]	6 [3, 9]
R6	1585	0.91 [0.63, 1.39]	9 [4, 11]	3.37 [2.06, 4.77]	5 [2, 10]
R7	3150	0.40 [0.30, 0.95]	1–6† [1, 9]	5.07 [4.03, 6.22]	12 [9, 12]
R9	3123	0.59 [0.40, 1.12]	1–6† [1, 9]	6.80 [5.71, 8.01]	15 [13, 15]
R10	1717	1.38 [0.79, 2.13]	12 [4, 14]	4.00 [2.88, 5.39]	8 [4, 11]
R11	2491	1.27 [0.76, 1.91]	11 [7, 13]	6.44 [5.25, 7.71]	14 [12, 15]
R12	4352	0.86 [0.47, 1.43]	8 [2, 11]	6.40 [5.39, 7.46]	13 [13, 15]
R13	2297	1.15 [0.76, 1.74]	10 [7, 12]	2.40 [1.42, 3.72]	2 [1, 5]
R14	3437	0.70 [0.42, 1.29]	7 [1, 10]	4.36 [3.41, 5.44]	10 [6, 12]
R15	2576	0.61 [0.40, 1.25]	1–6† [1, 10]	4.72 [3.60, 5.94]	11 [7, 12]
R16	1231	1.74 [1.16, 2.47]	13 [11, 14]	4.28 [3.06, 5.71]	9 [3, 13]

Point estimates with 95% percentile bootstrap CIs (B = 2,000, seed = 2026). Regions are listed by region number; Rank orders the partners by $\widehat{W}_1$ within each candidate effect modifier (1 = closest to Region 8); the bracketed rank interval is the 2.5th–97.5th percentile of the rank over 1,000 patient-level bootstrap replicates that resample within every region (Region 8 shared across partners; seed 20261011), and adjacent ranks whose intervals overlap are not resolved against each other. †Distances unresolved against their null floor (Section 4.2) are not ordered among themselves and share the point-rank range shown; the bracketed interval that follows it is that partner's own bootstrap rank interval. R6 and R5, adjacent in the SBP ranking, differ in $\widehat{W}_{1,\text{SBP}}$ by 0.003 mmHg, so their ranks are effectively tied.

4.4 | Clinical Interpretation

For each partner, $L^* = \Delta_{\text{clin}}/\widehat{W}_1$ at Δ_{clin} = 1%pt, with the percentile interval $[\Delta_{\text{clin}}/W_{1,U},\ \Delta_{\text{clin}}/W_{1,L}]$ of Section 2.4, is read against the illustrative bound L_{UB} for its effect modifier; on the W_1 scale of Figure 4 this is the position of each interval relative to the reference line at $\Delta_{\text{clin}}/L_{\text{UB}}$, with the limits read from Table 5 where an interval ends near the line; the line depends on the sponsor's illustrative bound and is not a threshold for pooling. An interval wholly below the line (L^* interval wholly above L_{UB}) means that even the upper confidence limit of the distance would require a slope above the bound to produce Δ_{clin}; an interval containing the line is compatible with a difference that could and one that could not produce it; an interval wholly above the line (L^* interval wholly below L_{UB}) means that a slope within the bound could produce Δ_{clin} even at the lower confidence limit of the distance. For distances unresolved against the null floor, only the upper confidence limit carries the reading. If the CATE were additionally assumed monotone, equation (5) would replace $\Delta_{\max}$ by the fraction $(W_1 + |\mu_1 - \mu_2|)/(2W_1)$ of it, which ranges from 0.66 to 1.00 across the age distances and from 0.72 to 1.00 across the SBP distances (point estimates, including the unresolved R2 on SBP).

5 | DISCUSSION

This paper answers the requirements set out in Section 2.1. W_1 meets requirement (i): in Study 1 the SMD is at chance at every sample size examined in the mean-matched cells, whereas W_1, like the KS statistic, reaches the discrimination target in every cell. It meets requirements (ii) and (iii): the heterogeneity bound $\Delta_{\max}$ (equation 7) translates distributional distance into a clinically interpretable scale, is sharp when the slope is the only restriction on the CATE and tightens under monotonicity (equation 5), and its constant is a clinical slope that the sponsor specifies. Its estimation is characterized by Study 2: for true distances of about 0.4 to 1.2 SD at $n \geq 100$ per region, percentile bootstrap coverage was below 0.95 beyond Monte Carlo error in 5 of the ten cells (Section 3.2.2).

These findings carry two implications for pooling assessments, illustrated by the GUSTO-I example (Section 4). First, when several candidate effect modifiers are considered, the per-modifier distances need not agree: a partner close to the anchor on one effect modifier can be distant on another, as in the contrast between R2 and R9. Second, within one effect modifier $L^* = \Delta_{\text{clin}}/W_1$ is a strictly decreasing function of W_1, so calibration does not reorder partners; it compares each partner's L^* with the bound L_{UB} specified for that modifier.

Across the 15 partners the orderings by age and by systolic blood pressure distance show no positive agreement (Spearman correlation −0.30, 95% interval −0.51 to 0.02 from the patient-level bootstrap of Table 5); the interval excludes the strong positive agreement under which one modifier's distances could stand in for the other's, so a single distance aggregated over the two modifiers could mask differences that a per-modifier assessment retains.

In GUSTO-I, whose regional differences are predominantly differences in location (partner standard deviations within a factor of 1.11 of the anchor's), W_1 ordered the 15 partners much as the SMD did (Spearman correlation 0.90, 95% interval 0.70 to 0.99, for age and 0.95, 0.90 to 0.99, for systolic blood pressure), and a registered survey of region pairs in GUSTO-I, IST-1 and IST-3 found agreement of the same order (Spearman correlation 0.839–0.996; repository).[29,30] The limitations of the SMD bind where distributional differences go beyond location—mixtures and rare displaced subgroups—and configurations beyond location cannot be excluded before the distances are computed. The reason to use W_1 is that, like the KS statistic, it is not blind where the SMD is, and that it carries a translation to the clinical scale whose constant is a per-unit clinical slope, valid for every CATE in the class and conservative for any particular one.

Because W_1 requires only baseline effect modifier distributions, regional data can be assembled from prior trials, registries, electronic health records, or other real-world evidence (RWE) sources, provided the source population approximates the population each region is expected to enrol, so that the assessment can be made at the planning stage. It uses neither treatment assignment nor outcomes, requires individual-level baseline values, and complements rather than replaces assessments based on observed treatment effects.

The W_1 distance is one input within a holistic regulatory evaluation.[3,4] It treats continuous effect modifiers marginally, leaving categorical and multivariate extensions for future work, and it is stated for collapsible effect scales (Section 2). Which baseline characteristics are effect modifiers is taken as given; their identification requires evidence on treatment-by-covariate interaction rather than on prognosis alone (Section 2.4) and lies outside the framework. Because W_1 is non-negative, $\widehat{W}_1$ exhibits positive bias near zero, zero percentile coverage when the true value is zero (S1), and undercoverage near the boundary and where the bias is large relative to the sampling standard deviation, with the intervals erring above the true value (Section 3.2.2).[31] The coverage reported for Study 2 from $n = 100$ per region refers to true distances of about 0.4 to 1.2 SD of the reference distribution, with equal regional sizes; the registered extension (Table 4) covers 0.2 SD at the application's sample sizes, unequal sizes, a heavy-tailed reference and grid-recorded data, but not every combination the application presents. Proposition 1 assumes a CATE common to all regions, so regional differences in extrinsic factors that alter τ itself lie outside what W_1 measures, and unmeasured effect modifiers may contribute to heterogeneity beyond what the distance captures.

The limitations of the calibration are those of the Lipschitz premise of Proposition 1. First, the constant is a clinical judgment (Section 2.4), so conclusions drawn from $\Delta_{\max}$, or from the comparison of L^* with L_{UB}, are conditional on it. Second, for a CATE that changes steeply over a narrow range of the effect modifier—a threshold or step function—the Lipschitz constant is large and equation (4) is correspondingly loose. This is by design: the bound is a worst case over the CATEs compatible with the stated L_{clinical}, and a loose bound overstates rather than understates the potential difference. Finally, the bound is per effect modifier. If the CATE is Lipschitz with respect to $d(x, y) = \sum_k L_k|x_k - y_k|$, then $|\bar{\tau}_1 - \bar{\tau}_2| \leq W_d(P_1, P_2)$, the Wasserstein distance between the joint distributions under cost d, and $W_d \geq \sum_k L_k W_1^{(k)}$; the per-modifier sum is a valid bound for a CATE additive across modifiers but, under interaction, can understate heterogeneity. Estimating W_d is outside this paper.

Inference tailored to the boundary behavior of $\widehat{W}_1$—resampling fewer than n observations[32] or the numerical delta method for directionally differentiable functionals[33]—could extend the operational range of the intervals at small distances.

The principal contribution of this paper is a distributional similarity measure for effect modifiers whose heterogeneity bound holds for every CATE with a given per-unit clinical slope, is sharp over that class and conservative for any particular CATE, and whose calibration constant is that slope. With bootstrap confidence intervals and clinical calibration via $\Delta_{\max}$ or L^*, W_1 supplies a quantitative input to the judgment of "similar enough", informing, not prescribing, pooling decisions.

AUTHOR CONTRIBUTIONS

[Author 1] conceived the research question and led the project. [Author 2] developed the mathematical framework and conducted the simulation study. [Author 3] contributed to the application example and manuscript preparation. All authors reviewed and approved the final manuscript.

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FINANCIAL DISCLOSURE

None reported.

CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

DATA AVAILABILITY STATEMENT

R code for computing the Wasserstein-1 distance and reproducing the simulation studies and the application is available at [repository URL]. The GUSTO-I individual patient data analysed in Section 4 are publicly available as the dataset `gusto` in the R package predtools (version 0.0.3; https://github.com/resplab/predtools); the trial is described in the original publication;[26] the analysis scripts load the data frame with `data(gusto, package = "predtools")` or from the identical CSV copy distributed with the code (N = 40,830 in both). Study 2 ran on a PSOCK cluster with L'Ecuyer–CMRG streams seeded per cell ($42 + 1{,}000k$, $k = 1, \ldots, 21$); replicates are assigned to workers dynamically, so the summaries are reproducible to Monte Carlo error, not bit for bit. A registered survey of region pairs in GUSTO-I, IST-1 and IST-3 is reported in Section efsec:discussion only as the range of its rank correlations, its tables not being reproduced; its protocol and outputs are included with the R code, as are the registered protocols of the Study 2 extension and of the Set 4 control check and their outputs.

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SUPPORTING INFORMATION

No supporting information accompanies this article; the R code, the registered protocols and all outputs are available as described in the Data availability statement.

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APPENDIX

A PROOFS AND MATHEMATICAL DETAILS

A.1 Upward Bias and Asymptotic Properties

Upward bias of $\widehat{W}_1$. Since $E\,\hat{F}_r(x) = F_r(x)$, Jensen's inequality gives $E\,|\hat{F}_1(x) - \hat{F}_2(x)| \geq |F_1(x) - F_2(x)|$ at every x, and integrating the non-negative integrand gives $E\,\widehat{W}_1 \geq W_1$ for all n_1, n_2.

The empirical Wasserstein-1 distance $\widehat{W}_1 = W_1(\hat{F}_1, \hat{F}_2)$ is a consistent estimator of $W_1(F_1, F_2)$; when $\int \sqrt{F(1-F)}\,dx < \infty$, $\sqrt{n}\,W_1(\hat{F}_n, F)$ converges in distribution to $\int |B(F(x))|\,dx$ with B a Brownian bridge.[34] For two samples with $n_1/(n_1+n_2) \to \lambda \in (0,1)$, when $\int \sqrt{F_r(1-F_r)}\,dx < \infty$ $(r = 1, 2)$ and $\{x : F_1(x) = F_2(x) \in (0,1)\}$ has Lebesgue measure zero, the directional derivative of W_1 at (F_1, F_2) is linear, the limit is Gaussian and the percentile bootstrap interval achieves asymptotically correct coverage;[16] this condition excludes $F_1 = F_2$, where the derivative is non-linear and the naive bootstrap is not consistent.[17,16] For an effect modifier recorded on a grid, the CDFs are step functions and the condition fails whenever they agree at any grid point inside the support; the finite-space results of Sommerfeld and Munk then apply,[18] with a Gaussian limit when the dual solution is unique. The condition concerns the population CDFs and cannot be checked from the samples, so for a distance unresolved against its null floor $F_1 = F_2$ is not excluded and the interval is read at its upper limit only.

A.2 Comparison with the Kolmogorov–Smirnov Bound; Attainment on the Outcome Scale

Neither bound dominates in general. On a support $[a, b]$, $W_1 \leq (b-a)\,D_{\mathrm{KS}}$ and $\mathrm{TV}(\tau) \leq L_{\mathrm{clinical}}\,(b-a)$, and $L_{\mathrm{clinical}} \cdot W_1 \leq \mathrm{TV}(\tau) \cdot D_{\mathrm{KS}}$ holds exactly when $W_1/D_{\mathrm{KS}} \leq \mathrm{TV}(\tau)/L_{\mathrm{clinical}}$. For a CATE whose slope has constant magnitude L_{clinical}, such as a linear CATE, $\mathrm{TV}(\tau) = L_{\mathrm{clinical}}(b-a)$ and equation (4) is never the looser of the two; the Kolmogorov–Smirnov bound gains as the CATE's slope falls below its maximum over much of the support, most sharply for a threshold CATE (Section 5).

The Kolmogorov–Smirnov bound is likewise sharp within its own class, by a step CATE of height $\mathrm{TV}(\tau)$ at the point where $|F_1 - F_2|$ is largest. A bound on the magnitude of τ itself tightens equation (4) further: on a support $[a, b]$ the attaining τ^* ranges over up to $L_{\mathrm{clinical}}(b-a)$, which on the risk-difference scale is 0.91 for age and 0.56 for systolic blood pressure over the recorded supports of Section 4, exceeding the highest 30-day mortality Lee et al. report for any age group (20.5% above 75 years[28]), beyond any baseline risk; for distributions with unbounded support no CATE bounded on the outcome scale attains the bound at all.